

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Engineered and deployed **Python-based feature pipelines** for enterprise ML applications across healthcare and finance verticals, integrating **SQL** data sources with **REST APIs** to power real-time fraud detection and identity verification systems; reduced pipeline latency by **40%** through asynchronous data orchestration and query optimization.
- Built and maintained **backend software services** using **Java Spring Boot** and **Python** supporting machine learning infrastructure; collaborated closely with ML teams and customers during pre-sales and post-sales engagement to gather requirements, design feature pipelines, and deliver production implementations with **99.5% uptime SLA**.
- Served as primary technical point of contact for customer onboarding and post-implementation support, conducting requirements analysis with both technical and non-technical stakeholders, translating business use cases into optimized ML feature pipelines, and recommending platform enhancements as customer needs evolved—resulting in **3 upsell opportunities** in final 6 months.
- Implemented **ML Ops infrastructure** including **CI/CD pipelines** via **Jenkins**, **Docker** containerization, and **AWS** cloud services (**EC2, S3, Lambda, CloudWatch**) to automate feature pipeline deployments across development, staging, and production environments; reduced deployment time by **55%**.
- Developed and debugged complex **SQL queries** across **PostgreSQL, MySQL, and Oracle** databases to validate data quality, design efficient schemas for feature stores, and support exploratory data analysis for ML feature engineering; conducted **code reviews** and **Git** collaboration practices to maintain code quality standards.
- Partnered with engineering and sales teams to interview and mentor junior engineers on backend software engineering best practices, ML infrastructure fundamentals, and customer-centric problem-solving; contributed to hiring pipeline documentation and technical onboarding materials for **4 new team members**.

SKILLS

Backend Software Engineering – Python, Java, Spring Boot, REST APIs, SQL, PostgreSQL, MySQL, Oracle.

Machine Learning Operations & Infrastructure – ML feature pipelines, data pipeline orchestration, feature stores, model serving infrastructure, asynchronous processing.

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git, CI/CD automation, Linux.

Data Engineering & Analytics – SQL optimization, schema design, data validation, exploratory data analysis, pandas, NumPy.

Customer & Stakeholder Engagement – requirements gathering, technical onboarding, post-implementation support, cross-functional collaboration, problem-solving, storytelling.

Data Science & Statistics – scikit-learn, statistical analysis, hypothesis testing, supervised and unsupervised learning, model evaluation metrics.

Business & Domain Knowledge – healthcare data pipelines, financial services ML applications, fraud prevention systems, identity verification, recommendation systems.

EDUCATION AND TRAINING

PROJECTS

Real-Time Feature Pipeline for Fraud Prevention at Scale (HCL Technologies)

- Architected and deployed a **Python-based feature pipeline** processing **10M+ financial transactions daily** across multiple banking customers, integrating **SQL** data warehouses with **REST APIs** to compute real-time fraud risk scores for credit card authorization and identity verification use cases.
- Implemented **ML Ops infrastructure** using **Docker** containerization, **AWS Lambda** for serverless execution, **S3** for feature storage, and **Jenkins** CI/CD pipelines to automate feature computation, validation, and model inference; reduced end-to-end latency from **8 seconds to 2 seconds** per transaction.
- Designed **SQL schemas** across **PostgreSQL and Oracle** to optimize feature materialization, implemented **data validation** workflows to ensure **99.8% data quality**, and collaborated with customer ML teams to iterate on feature definitions; reduced false-positive fraud alerts by **32%** while maintaining **98% true-positive detection rate**.
- Gathered requirements from healthcare and finance domain stakeholders (non-technical operations teams and technical data scientists), translated use cases into implementable feature specifications, and provided ongoing post-sales technical support across **5 customer deployments** spanning cancer risk scoring and cross-border payment fraud detection.
- Built monitoring and alerting via **CloudWatch** dashboards and custom **Python** logging to track pipeline performance, feature freshness, and model drift; enabled **sub-10-minute incident response** for production SLA breaches.

Machine Learning Feature Platform Integration & Customer Onboarding (HCL Technologies)

- Led end-to-end technical onboarding for **3 new enterprise customers** adopting the Chalk feature platform, conducting requirements workshops with both technical and non-technical stakeholders to understand healthcare recommendation systems, financial fraud prevention, and identity verification use cases.
- Engineered **REST API integrations** connecting customer ERP systems and data warehouses to Chalk's feature platform using **Java Spring Boot** microservices, enabling automated feature computation and serving with **sub-100ms latency**.
- Recommended and implemented ML pipeline enhancements including **feature store optimization**, **model inference caching**, and **data lineage tracking**, resulting in **3 upsell opportunities** and expanded contracts worth **\$250K+ ARR**.
- Collaborated with sales teams during pre-sales technical due diligence to validate customer data readiness, design proof-of-concept feature pipelines, and present architecture recommendations; authored technical specification documents for **5 customer projects**.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)